

Isolation of Multiple Lipoprotein [a] Charge Forms in Human Plasma by Liquid Phase Isoelectric Focusing

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Introduction

Lipoprotein [a] (Lp[a]) resembles low density lipoprotein (LDL) in that both lipoproteins contain apoB, a 514 kDa hydrophobic apolipoprotein which transverses the phospholipid outer layer of the lipid core of these molecules. In addition to apoB, Lp[a] also contains apo[a], a highly glycosylated apolipoprotein which is disulfide-linked to apoB. Genetically-determined size variations of apo[a] ranging from 420-830 kDa have been reported.¹ While the contribution of apo[a] to the whole Lp[a] particle (3.8×10^6 Da) makes it distinguishable from LDL in size separations such as gel filtration and non-denaturing gradient gel electrophoresis, post-translational modifications such as glycosylation might generate multiple charge forms that would not be resolved using these techniques.

As an alternative, we consider separation of Lp[a] subclasses on the basis of charge. Liquid phase isoelectric focusing (LP-IEF) seems ideally suited for the charge fractionation of multi-million dalton lipoprotein particles such as Lp[a] and LDL. Unlike conventional IEF which is performed in polyacrylamide or agarose matrices, LP-IEF is performed in free solution. This is advantageous since a charged species will exhibit free mobility relative to its net charge in solution. In contrast, mobility is restricted in gel matrices, even when the lowest practical total monomer concentration (%T) is used, and large macromolecules characteristically exhibit high retardation coefficients. Therefore, large proteins should focus more rapidly in free solution than in gel, and time-dependent decay of the pH gradient is less likely to effect the separation.

Methods

Sample Preparation

Blood was drawn from fasting, normal individuals into 20 ml syringes containing 3 mM disodium ethylenediaminetetraacetic acid (Na_2 EDTA). Following centrifugation for 10 minutes at 1,500 g, the resulting plasmas were prestained with Sudan Black B (SBB) as previously described.² Briefly, 8 ml of plasma was mixed with 2 ml of a saturated solution of SBB in 100% ethylene glycol and incubated 12 hours at 4 °C. Prior to use, prestained plasma was centrifuged 10 minutes at 1,500 g to pellet any precipitated stain.

Typically, 7 ml plasma was diluted 1:5 in 20% ethylene glycol containing 4% Bio-Lyte® ampholytes (pH range 3–10). Samples were filtered through a 1.2 mm filter and applied to the Rotofor® preparative IEF cell.

Running Conditions

LP-IEF was performed at 10 W constant power at 4 °C. Initial conditions were 280 V and 38 mA, and equilibrium was reached in 5 hr at 720 V and 12 mA. Fractions were collected and the pH of each was measured (Figure 1). Lipoprotein-enriched fractions, as determined by O.D._{507 nm} (SBB absorbance maximum), were pooled and refractionated over a pH gradient of a more specific range. For this, the volume was readjusted to 35 ml with 20% ethylene glycol. No additional ampholytes were added. Upon refractionation at 6–8 W constant power at 4 °C, initial conditions were 610 V and 16 mA and equilibrium was reached in 4 hr at 1,120 V and 5 mA. Fractions were collected and the pH of each was measured (Figure 1).

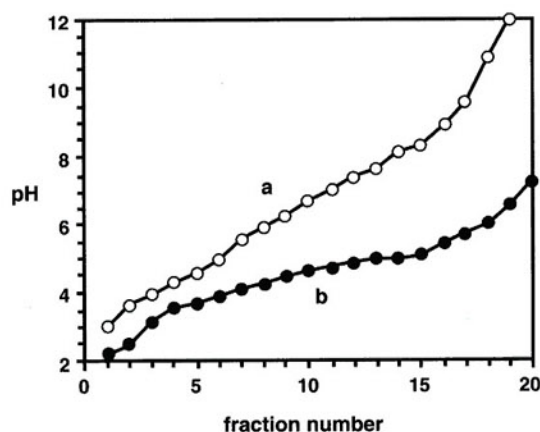


Fig. 1. pH gradients formed during LP-IEF. A. First fractionation of human plasma using 4% Bio-Lyte ampholytes (pH range 3–10). **B.** Fractions 1–10 were pooled and refractionated without added ampholytes.

Analytical Procedures

Prior to analyses, fractions were dialyzed against 60 mM Tris-HCl pH 6.8 to remove ampholytes. Lp[a] was quantified by radioimmunoassay for apo[a].³ Distribution of lipoproteins in SBB-prestained plasmas was shown by non-denaturing gel electrophoresis in piperazine diacrylamide (PDA)-crosslinked 3–20% concave exponential polyacrylamide gradients. Bands could be visualized without additional staining and were inten-

sified by semi-dry electrotransfer to polyvinylidene difluoride (PVDF) membrane (Figure 2A). Charge forms of Lp[a] were demonstrated by electrophoresis on agarose gel film (Ciba-Corning Diagnostics) followed by transfer to nitrocellulose membrane. Nitrocellulose was sequentially immunoprobed with goat anti-human apo[a] and 125 I-labeled rabbit anti-goat IgG and visualized by autoradiography (Figure 2B).

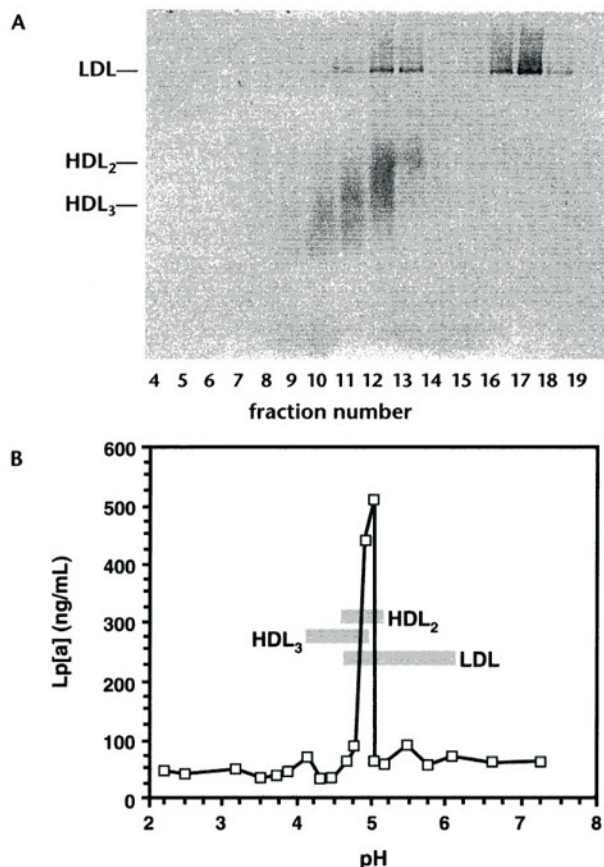


Fig. 2. Distribution of lipoproteins in human plasma as separated by LP-IEF. **A.** nondenaturing gradient electrophoresis of fractions 4–19 showing distribution of HDL₂, HDL₃, and two major charge forms of LDL. Lp[a] concentrations were too low to be detected by SBB. **B.** Quantitation of Lp[a] by radioimmunoassay. The overlap of HDL and LDL with Lp[a] peak fractions is indicated.

Results and Discussion:

Four charge forms of Lp[a] were isolated from a single human plasma sample as exemplified by agarose electrophoresis (Figure 3). Agarose gel films provide a relatively non-restrictive media in which proteins migrate essentially according to their net charge. Theoretically, relative electrophoretic mobility (REM) should decrease proportionally as isoelectric point (as estimated from the pH of each fraction) is increased. Here,

we observed an increase in REM suggesting a continuum of Lp[a] charge forms possibly due to varying degrees of glycosylation. A correlation between the REM of *gs-I* through *IV* and the pH at which each was focused was indicated by linear regression analysis where $r = 0.992$. This is in agreement with previous studies which have shown that Lp[a] mobility is decreased by enzymatic deglycosylation.⁴

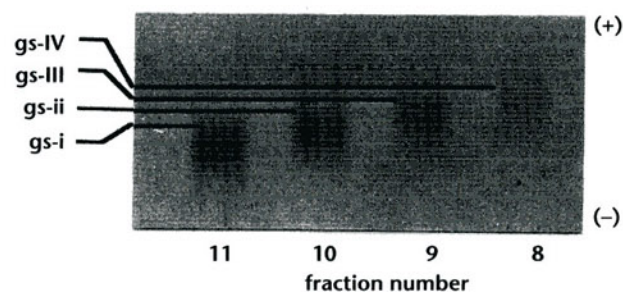


Fig. 3. Agarose gel film electrophoresis of fractions 8–11 followed by immunoblotting. Four charge forms of Lp[a], designated *gs-I* through *IV* in order of increasing electrophoretic mobility, were detected in a single human plasma. REM is expressed as mobility relative to the mobility of *gs-I*.

In conclusion, multiple charge forms of Lp[a] were resolved by LP-IEF. Since differential atherogenicity of Lp[a] subclasses could explain the relationship between plasma Lp[a] levels and the risk of atherosclerotic clinical events, isolation of specific Lp[a] subclasses could provide the opportunity to clarify the underlying mechanisms responsible for disease events. Further experimentation is required to determine how apo[a] isoform size relates to post-translational modification such as glycosylation. Finally, we propose that preparative LP-IEF might be used as the initial step in a purification scheme designed to isolate lipoprotein subclasses from human plasma.

References

1. Gaubatz, J. W., Ghanem, K. I., Guevara, J., Nava, M. L., Patsch, W. and Morrisett, J. D., *J. Lipid Res.*, **31**, 603-613 (1990).
2. Mills, G. L., Lane, P. A. and Weech, P. K., in *Guidebook to Lipoprotein Technique*, Elsevier, Amsterdam, 192-193 (1984).
3. Pepin, J. M., O'Neil, J. A. and Hoff, H. F., *J. Lipid Res.*, **32**, 317-327 (1991).
4. Yashiro, A., O'Neil, J. A. and Hoff, H. F., *J. Biol. Chem.*, submitted (1992).

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